CPPC 18/L

COMPUTER NETWORK & SECURITY

ACTIVITY No.4

DHCP IPv4 Configuration

Submitted by:

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BS CpE 3B

Submitted to:

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Subject Professor

Activity 4: DHCP IPv4 Configuration

I. Objectives

- To configure and turn on the DHCP service on a server.
- To allow PCs to automatically receive IPv4 addresses.
- To understand how DHCP helps assign IP addresses without manual configuration.
- To verify if the client PCs can successfully obtain IP addresses from the DHCP server.

II. Materials

- Cisco Packet Tracer
- 1 Server
- 1 Switch
- 2–3 PCs
- Laptop
- Copper Straight-Through cables

III. Procedures

1. Create the Network Topology

- Open Cisco Packet Tracer.
- Add the required devices in the workspace:
  - 1 Server
  - 1 Switch
  - 2–3 PCs
- Connect all devices using Copper Straight-Through cables:
  - PC to Switch
  - Server to Switch

- Make sure that all connections are properly connected and active.

2. Configure the Server IP Address

- Click the Server.
- Go to Desktop.
- Open IP Configuration.
- Set the server with the following IP configuration:
  - IP Address: 192.168.1.1
  - Subnet Mask: 255.255.255.0
- This allows the server to become part of the network and provide IP addresses to the PCs.

3. Enable DHCP Service

- Click the Server.
- Go to the Services tab.
- Select DHCP.
- Turn the DHCP service ON.
- This enables the server to automatically assign IPv4 addresses to the client computers.

4. Configure the DHCP Pool

- In the DHCP settings, set up the DHCP pool.
- Enter the following information:
  - Pool Name: IPv4
  - Default Gateway: 0.0.0.0
  - DNS Server: 0.0.0.0
  - Start IP Address: 192.168.1.1

- Subnet Mask: 255.255.255.0
- Maximum Number of Users: 10

- Click Add to save the DHCP pool.
- This pool will be used to assign available IP addresses to the connected PCs.

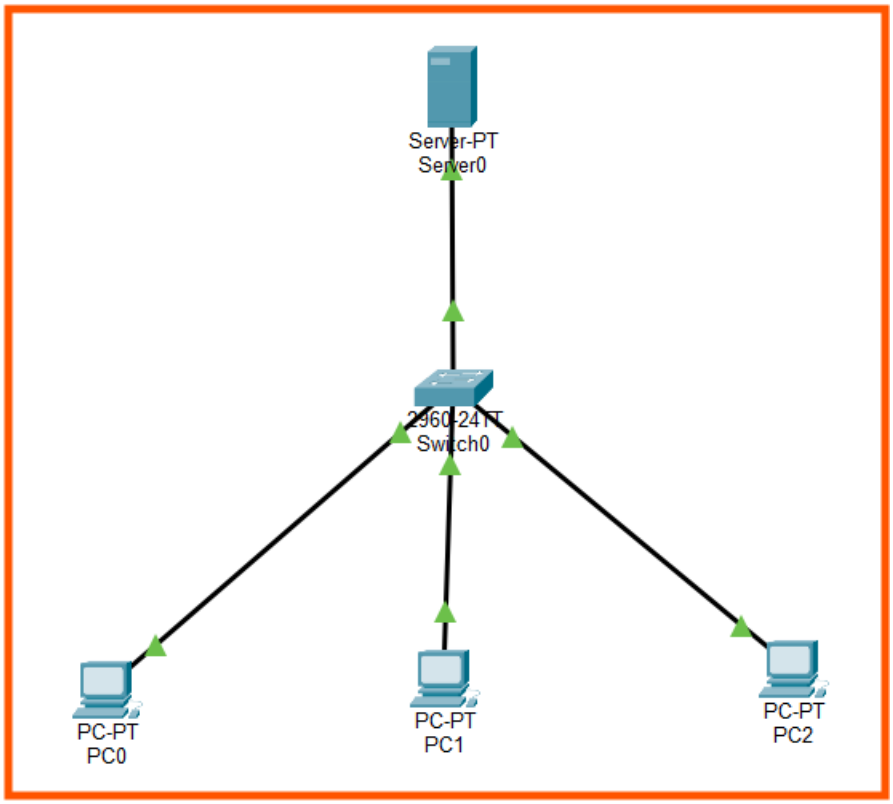
5. Configure PCs to DHCP

- Click each PC one at a time.
- Go to Desktop.
- Open IP Configuration.
- Select DHCP.
- Wait until the message "DHCP request successful" appears.
- Repeat the same steps for the remaining PCs.
- Each PC should automatically receive an IPv4 address from the server.

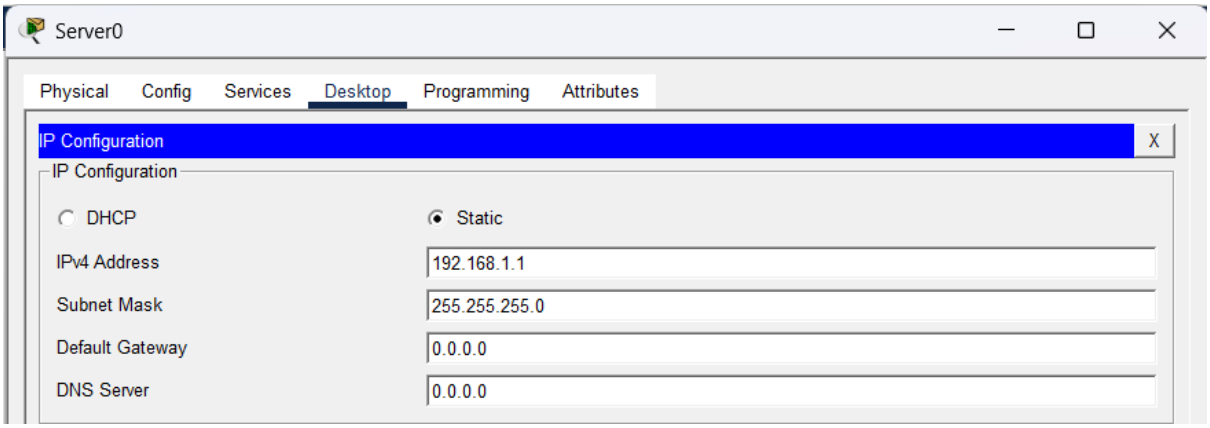
IV. Conclusion

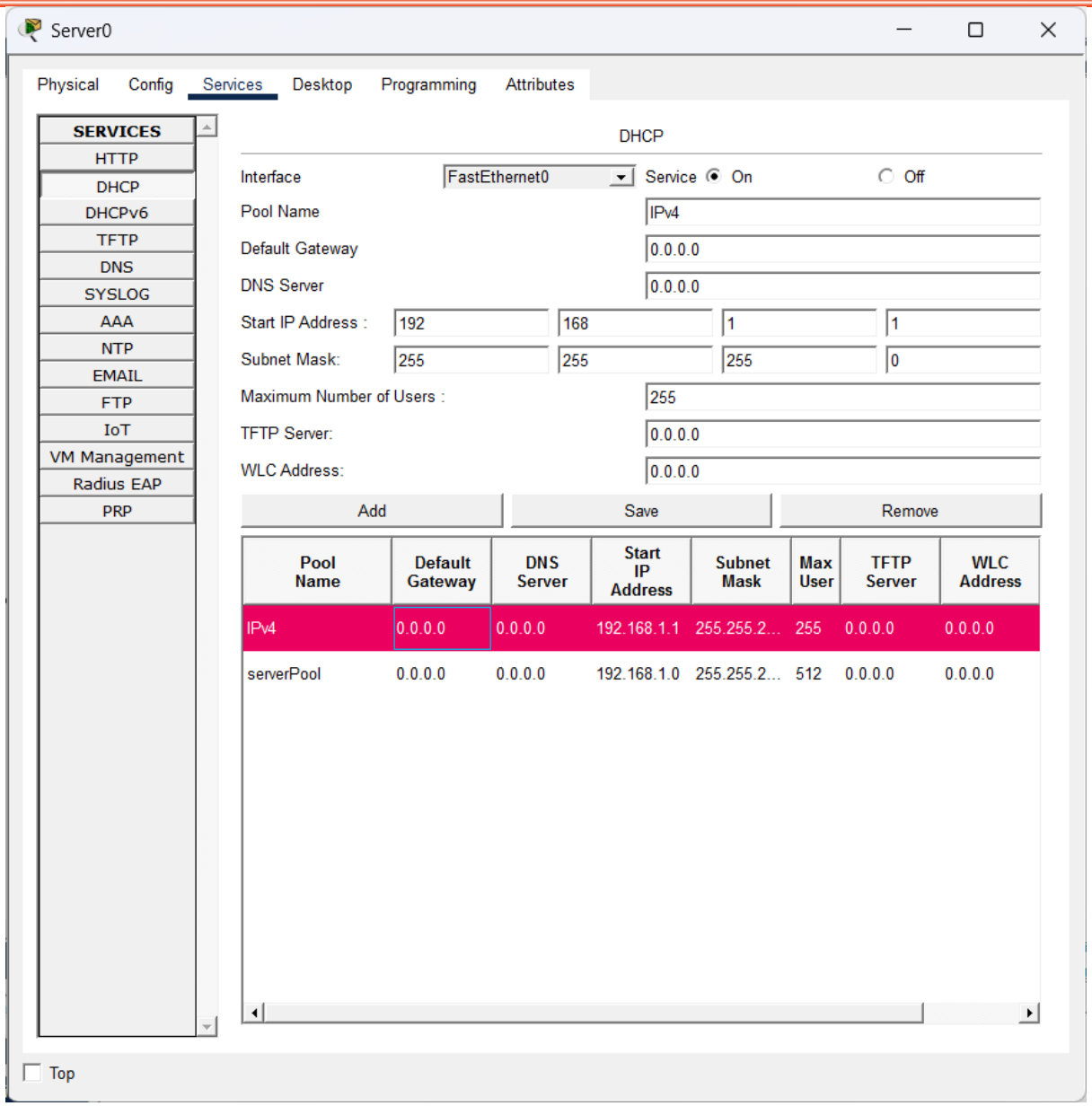
The activity showed how DHCP IPv4 can be configured in Cisco Packet Tracer by enabling the DHCP service on a server and setting up a DHCP pool. Through this configuration, the connected PCs were able to automatically receive valid IP addresses without manually assigning them one by one. This proves that DHCP makes network setup easier and more efficient, especially when multiple devices are connected to the same network. The successful DHCP request from the PCs also confirmed that the server was properly configured and that the DHCP service was working correctly.

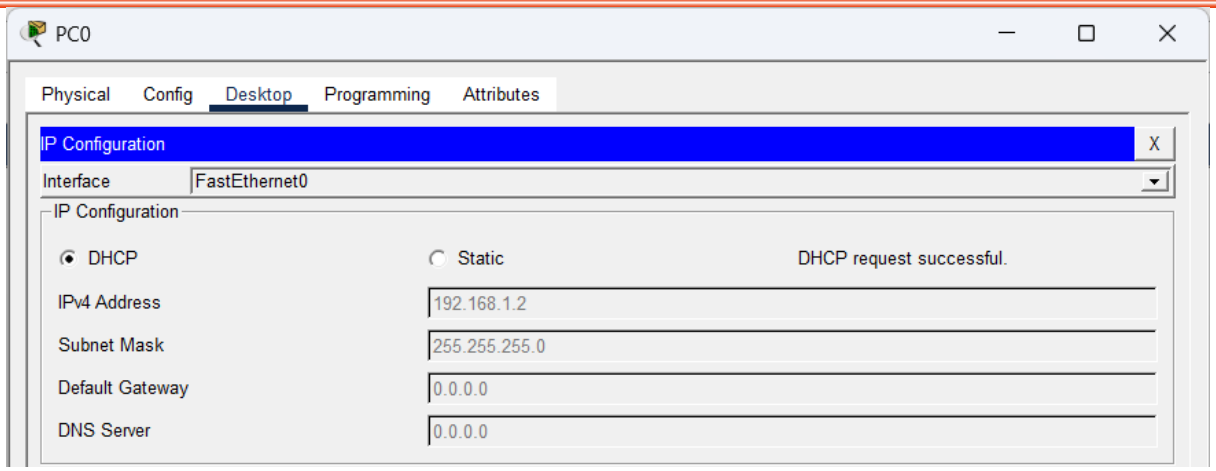
V. Documentation

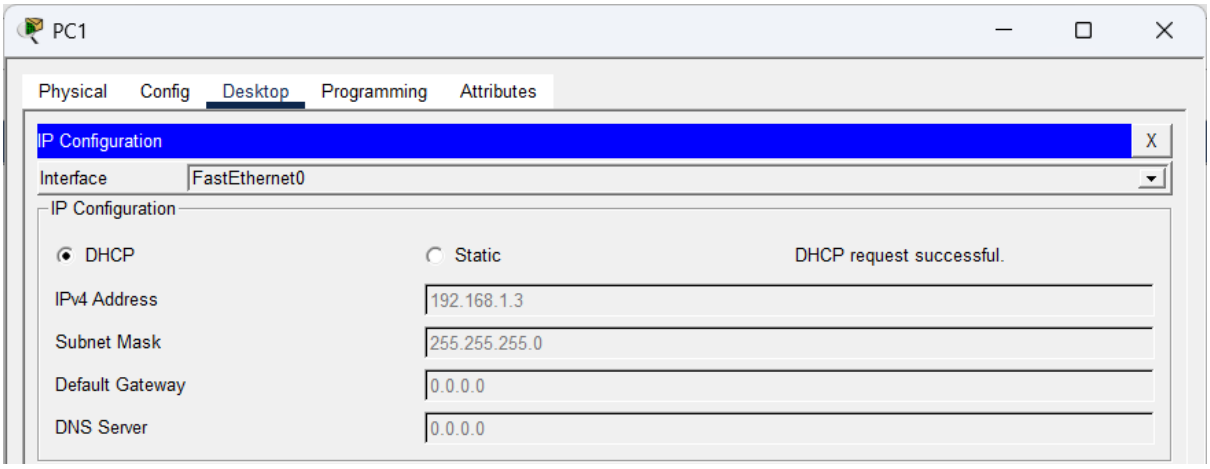


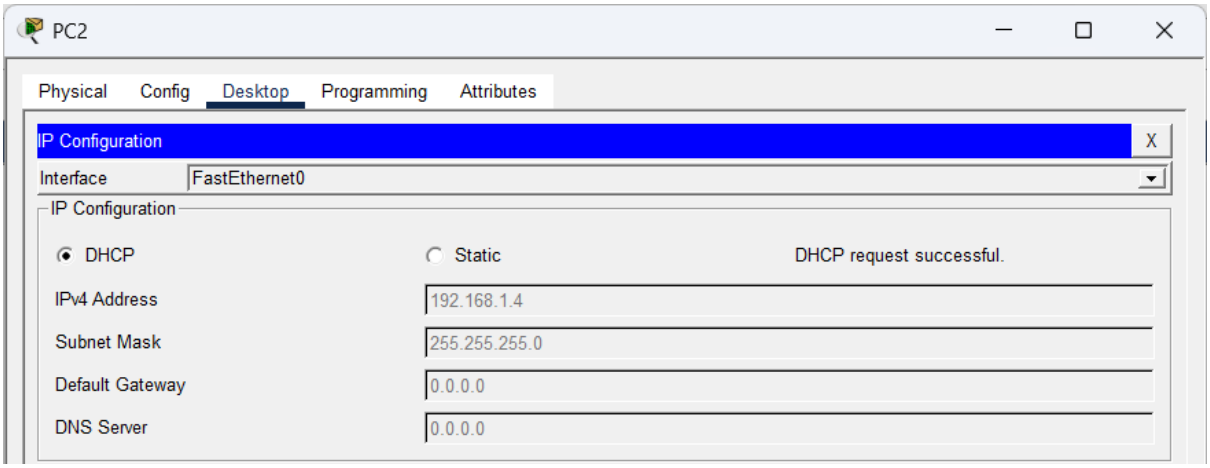
Server




Server0
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP
DHCP
Interface
FastEthernet0
Service
On
Off
Pool Name
IPv4
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
Start IP Address :
192
168
1
1
Subnet Mask:
255
255
255
0
Maximum Number of Users :
255
TFTP Server:
0.0.0.0
WLC Address:
0.0.0.0
Add
Save
Remove
Pool Name
Default Gateway
DNS Server
Start IP Address
Subnet Mask
Max User
TFTP Server
WLC Address
IPv4
0.0.0.0
0.0.0.0
192.168.1.1
255.255.2...
255
0.0.0.0
0.0.0.0
serverPool
0.0.0.0
0.0.0.0
192.168.1.0
255.255.2...
512
0.0.0.0
0.0.0.0
Top


PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
DHCP request successful.
IPv4 Address
192.168.1.2
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0


PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
DHCP request successful.
IPv4 Address
192.168.1.3
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0


PC2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
DHCP request successful.
IPv4 Address
192.168.1.4
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0